

CS1005 Discrete Structures

Thursday, November 30, 2023

Course Instructor

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Serial No:

2nd Mid Term Exam

Total Time: 1 Hour

Total Marks: 65

Signature of Invigilator

Roll No

Section

Signature

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Verify at the start of the exam that you have a total of **six (06)** questions printed on **-- (--)** pages including this title page.
2. Attempt all questions on the question-book and in the given order.
3. The exam is closed books, closed notes. Please see that the area in your threshold is free of any material classified as 'useful in the paper' or else there may a charge of cheating.
4. Read the questions carefully for clarity of context and understanding of meaning and make assumptions wherever required, for neither the invigilator will address your queries, nor the teacher/examiner will come to the examination hall for any assistance.
5. Fit in all your answers in the provided space. You may use extra space on the last page if required. If you do so, clearly mark question/part number on that page to avoid confusion.
6. Use only your own stationery and calculator. If you do not have your own calculator, use manual calculations.
7. Use only permanent ink-pens. Only the questions attempted with permanent ink-pens will be considered. Any part of paper done in lead pencil cannot be claimed for checking/rechecking.

	Q-1	Q-2	Q-3	Q-4	Q-5	Q-6	Total
CLO(s)	03	03	03	03	03	03	
Total Marks	10	10	15	10	10	10	65
Marks Obtained							

Vetted By: _____ **Vetter Signature:** _____

University Answer Sheet Required: No ☐ Yes ☐

Question No 01: [2 * 5 = 10]

1. If each element of set P is also an element of set Q, then P is said to be **Subset** of Q and Q is said to be **Superset** of P.
2. In a group of 70 people, 37 like coffee, 52 like tea and each person likes at least one of the two drinks. How many people like both coffee and tea **19**
3. According to combination rule, if the total number of outcomes are 'r' and distinct outcome collection is 'n' then combinations are calculated as **$C(n,r)=n!/(n-r)! r!$**
4. The relation $A \times A$ is known as **Universal Relation**
5. The transitive closure of the relation $\{(0,1), (1,2), (2,2), (3,4), (5,3), (5,4)\}$ on the set $\{1, 2, 3, 4, 5\}$ is **$\{(0,2)\}$** .

QUESTION 2: In a class of 40 students, 20 have chosen Mathematics, 15 have chosen mathematics but not biology. If every student has chosen either mathematics or biology or both, find the number of students who chose both mathematics and biology and the number of students chose biology but not mathematics. Draw Venn diagram. [10]

Let, $M \equiv$ Set of students who chose mathematics

$B \equiv$ Set of students who chose biology

$$n(M \cup B) = 40$$

$$n(M) = 20$$

$$n(B) = n(M \cup B) - n(M)$$

$$\Rightarrow n(B) = 40 - 20 = 20$$

$$n(M - B) = 15$$

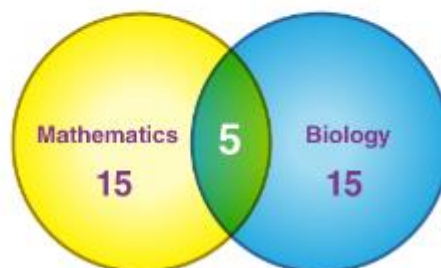
$$n(M) = n(M - B) + n(M \cap B)$$

$$\Rightarrow 20 = 15 + n(M \cap B)$$

$$\Rightarrow n(M \cap B) = 20 - 15 = 5$$

$$n(B - M) = n(B) - n(M \cap B)$$

$$\Rightarrow n(B - M) = 20 - 5 = 15$$

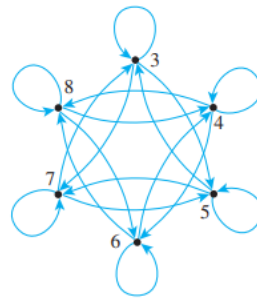


QUESTION 3: Let $A = \{3, 4, 5, 6, 7, 8\}$ and define a relation R on A as follows. [15]

a) Find the Relation R , For all $x, y \in A$, $x R y \Leftrightarrow 2 \mid (x - y)$. [5] & directed graph.

Solution Note that $3 R 3$ because $3 - 3 = 0$ and $2 \mid 0$ since $0 = 2 \cdot 0$. Thus there is a loop from 3 to itself. Similarly, there is a loop from 4 to itself, from 5 to itself, and so forth, since the difference of each integer with itself is 0, and $2 \mid 0$.

Note also that $3 R 5$ because $3 - 5 = -2 = 2 \cdot (-1)$. And $5 R 3$ because $5 - 3 = 2 = 2 \cdot 1$. Hence there is an arrow from 3 to 5 and also an arrow from 5 to 3. The other arrows in the directed graph, as shown below, are obtained by similar reasoning.



c) Is relation a equivalence relation or partial relation, and provide your reasoning?[5]

The relation is equivalence relation.

Question No 04: For all sets A and B , Prove that $A - (A \cap B) = A - B$ with the help of Set identities laws. (Please ensure that you mention the name of every law you apply at each step. Failure to do so will result in a deduction of marks for the question). [10]

Construct an algebraic proof that for all sets A and B ,

$$A - (A \cap B) = A - B.$$

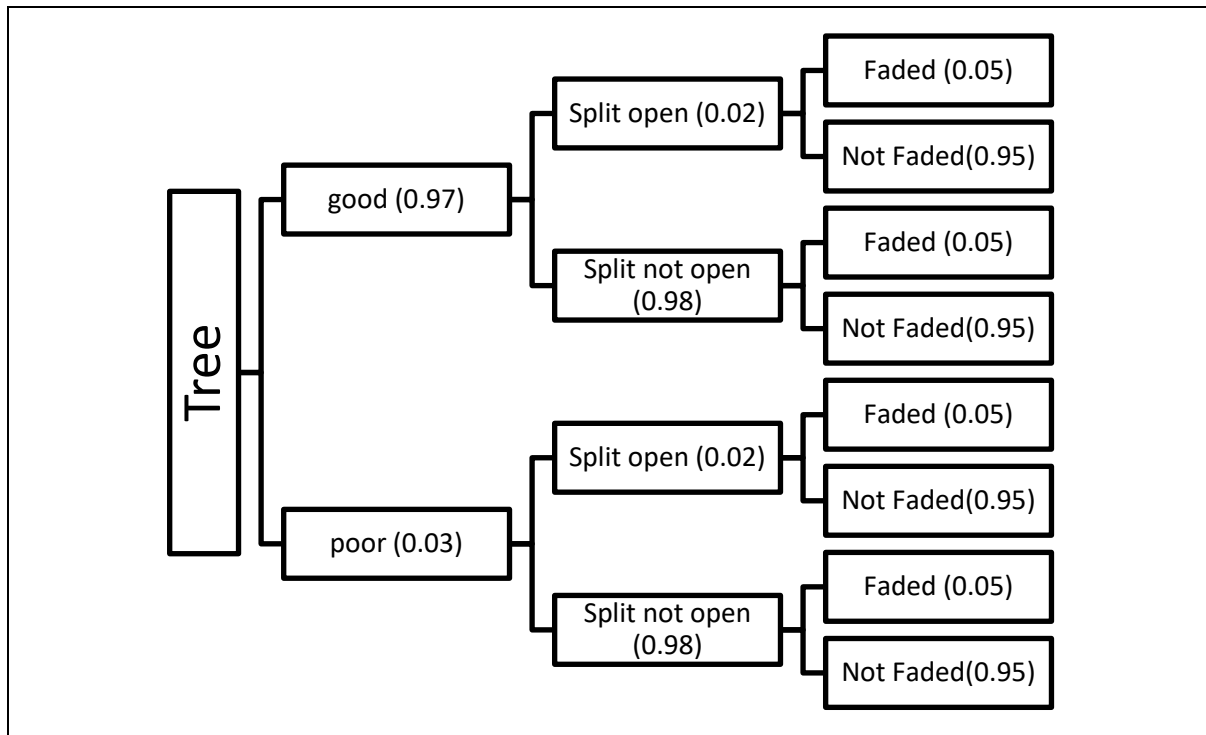
Cite a property from Theorem 6.2.2 for every step of the proof.

Solution Suppose A and B are any sets. Then

$$\begin{aligned} A - (A \cap B) &= A \cap (A \cap B)^c && \text{by the set difference law} \\ &= A \cap (A^c \cup B^c) && \text{by De Morgan's laws} \\ &= (A \cap A^c) \cup (A \cap B^c) && \text{by the distributive law} \\ &= \emptyset \cup (A \cap B^c) && \text{by the complement law} \\ &= (A \cap B^c) \cup \emptyset && \text{by the commutative law for } \cup \\ &= A \cap B^c && \text{by the identity law for } \cup \\ &= A - B && \text{by the set difference law.} \end{aligned}$$

QUESTION 5: A manufacturer carried out a survey of the defects in their soft toys. It is found that the probability of a toy having poor stitching is 0.03 and that a toy with poor stitching probability of 0.7 of splitting open. A toy without poor stitching has a probability of 0.02 of splitting open. The manufacturer also found that soft toy become faded with the probability of 0.05 and that this defect is common for poor stitching and splitting open. [10]

1. Construct the possibility tree showing all possible outcomes of this experiment and compute their event probabilities. [6]



2. Find the probability that the soft toys have none of these 3 defects. [4]

Soft toys does not have ay three defects = $(0.97) * (0.98)* (0.95)$

$$= 0.90307$$

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QUESTION 6: Hexadecimal numbers are made using the sixteen digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F. They are denoted by the subscript 16. For example, $9A2D_{16}$ and $BC54_{16}$ are hexadecimal numbers. [10]

1. How many hexadecimal numbers begin with one of the digits 3 through B, end with one of the digits 5 through F, and are 5 digits long? [5]

First digit from 3 to B = 11 choices

Last digit from 5 to F = 9 choices

Total digits = 5

$$\begin{aligned} &= 11 * 16 * 16 * 16 * 9 \\ &= 405504 \end{aligned}$$

2. How many hexadecimal numbers begin with one of the digits 4 through D, end with one of the digits 2 through E, and are 6 digits long? [5]

First digit from 4 to D = 10 choices

Last digit from 2 to E = 13 choices

Total digits = 6

$$\begin{aligned} &= 10 * 16 * 16 * 16 * 16 * 13 \\ &= 8519680 \end{aligned}$$

GOOD LUCK 😊